KRISHNA BIRANJE

Aspiring Mechatronics Engineer

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Manipal, India

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SUMMARY

I am an aspiring Mechatronics Engineer with a strong foundation in **robotic systems and actuation design**. My experience encompasses **CAD modelling**, **FEA-driven simulations**, and leading teams to deliver **innovative robotic solutions**. I excel in **prototype development** and thrive in competitive environments, always aiming to push the boundaries of technological performance in my projects.

EXPERIENCE

Senior Mechanical & Robotics Engineer

Mars Rover Manipal

10/2023 - 08/2025 Manipal, India

Worked on the design and development of robotic systems within a competitive team environment.

- Designed and **developed a 6-DOF robotic manipulator** using **CAD modelling, link geometry optimisation, joint configuration, actuator selection, torque calculations, and kinematic analysis.**
- Performed finite element modelling (ANSYS Workbench), **including static, dynamic, and fatigue analyses, to validate stress distribution, stiffness, factor of safety,** and the structural integrity of manipulator and drivetrain components.
- **Designed and analysed a high-reduction cycloidal drive (50:1)**, conducting **contact analysis, stress distribution assessment, and stiffness evaluation** of cycloidal discs and load-bearing components.
- Engineered a lightweight, **terrain-adaptive 3D-printed wheel assembly**, applying **material characterisation**, compliance tuning, and **design-for-additive-manufacturing** principles while optimising infill, layer height, wall thickness, and print orientation for stiffness, durability, and terrain performance.

Head Of Operations

Mars Rover Manipal

10/2025 Manipal, India

Leading a competitive robotics team focused on autonomous rover design and development

- Leading a team of 50 undergraduate engineers to **design and build competition-grade autonomous rovers for international challenges.**
- Streamlined debugging and validation workflows, reducing overall testing time by approximately 20%.
- **Managing project planning**, finances, logistics, and sponsorship outreach to ensure sustainable team operations.

EDUCATION

Bachelor of Technology in Mechatronics Engineering

Manipal Institute of Technology

08/2023 - 09/2027 Manipal, India

KEY ACHIEVEMENTS

Awards in Robotics Competitions

SpaceCert 2025 – **European Space Foundation** – Senior Mechanical Engineer

1st Place – International Rover Challenge 2025 (50+ teams)

10th Place – **European Rover Challenge 2025** (100+ teams)

SKILLS

FEA | CAD | Robotics | ANSYS Workbench | SolidWorks | ANSYS | 3D Printing | CNC | Precision Tooling | Material Characterization | Rapid Prototyping | Actuators

PROJECTS

6 DOF Robotic Manipulator

01/2023 - 05/2023 Manipal, India

Project focused on creating a versatile robotic manipulator

- Developed a 6-DOF robotic manipulator with optimised link geometry and actuator layout **to maximise workspace reach and payload capacity.**
- Designed **a custom end effector enabling continuous rotation for robotic manipulation tasks.**
- Conducted finite element analysis in **ANSYS** Workbench to verify performance under static and dynamic loading conditions.

Cycloidal Drive Transmission

06/2023 - 09/2023 Manipal, India

Project focusing on the development of a cycloidal drive transmission

- Developed a high-reduction cycloidal drive (50:1) for robotic manipulation applications **requiring high output torque in a compact form factor.**
- Conducted detailed finite element analysis on cycloidal discs and load-bearing components to assess stress distribution and stiffness under operating loads.

3D-Printed Wheel Assembly

09/2023 - 12/2023 Manipal, India

Project on creating a 3D-printed wheel assembly for robotics

- Developed a lightweight, terrain-adaptive wheel capable of traversing uneven surfaces while **maintaining traction and accommodating structural loads** within a **3D-printed assembly**.
- Designed a **WEGS-inspired wheel geometry** and **evaluated multiple material options** while iteratively **tuning 3D printing parameters.**